

OEM O&M; MANUAL (EXTRACT)

Deccan Flow Systems DFS-CC-250 | Type DFS-3300 Mechanical Seal

Extract from the Operation & Maintenance Manual for Deccan Flow Systems model **DFS-CC-250** centrifugal pump fitted with **Type DFS-3300** cartridge mechanical seal. Applies to pump tag P-1101A / P-1101B.

Section 6 — Mechanical Seal & Seal Flush

6.1 The DFS-3300 cartridge seal requires a clean, cool flush. For hot hydrocarbon service the pump **must be operated on API Plan 52** with the buffer-fluid reservoir maintained between the MIN and MAX marks at all times. Operating the seal with the reservoir below MIN, or with the cooling coil isolated, will cause buffer-fluid coking and **premature seal face failure, typically within 8–12 weeks**.

6.2 Buffer fluid: ISO VG 32 mineral oil. Replace every 4000 running hours or on discolouration. Cooling-water flow to the reservoir coil: min 1.5 m³/hr.

Section 7 — Shaft Alignment

7.1 Pump-to-motor alignment **must be checked by laser** after every seal change and after any disturbance to the baseplate. Permissible misalignment:

Type	Acceptable	Maximum (correct before start)
Parallel offset	≤ 0.05 mm	0.10 mm
Angular	≤ 0.04 mm/100mm	0.08 mm/100mm

7.2 **Running a DFS-3300 seal with misalignment above the maximum is the single most common cause of repeat seal failure.** Misalignment imposes cyclic load on the seal faces and bearings and will not be resolved by seal replacement alone — the alignment must be corrected.

Section 8 — Torque & Reassembly

Fastener	Size	Torque (Nm)
Seal gland nuts	M12	45
Casing cover bolts	M20	210
Coupling hub	-	Hydraulic, per 8.4